DH7512

Receiving Card

V1.1.0

Specifications

西安诺瓦星云科技股份有限公司

Introduction

The DH7512 is a general receiving card developed by NovaStar. A single DH7512 loads up to 512×512 pixels (NovaLCT V5.3.1 or later required). Supporting color management, 18Bit+, pixel level brightness and chroma calibration, individual Gamma adjustment for RGB, and 3D functions, the DH7512 can greatly improve the display effect and user experience.

The DH7512 has 12 HUB75 output to modules. It supports up to 24 groups of parallel RGB data. The DH7512 has improved electromagnetic compatibility and is suitable to various on-site setups.

Features

Improvements to Display Effect

- Color management
  Switch the color gamut of the screen between multiple gamuts to enable more precise colors on the screen.
- 18Bit+
  Improve the LED display grayscale by 4 times to avoid grayscale loss due to low brightness and allow for a smoother image.
- Pixel level brightness and chroma calibration
  Working with NovaLCT and NovaCLB, the receiving card supports brightness and chroma calibration on each LED, which can effectively remove color discrepancies and greatly improve LED display brightness and chroma consistency, allowing for better image quality.
- Quick adjustment of dark or bright lines
  The dark or bright lines caused by splicing of cabinets or modules can be adjusted to improve white balance offset, allowing for a more realistic image.
- Image rotation in 90° increments
  The display image can be set to rotate in multiples of 90° (0°/90°/180°/270°).
- 3D function
  Working with the independent controller which supports 3D function, the receiving card supports 3D image output.
- Individual Gamma adjustment for RGB Working with NovaLCT (V5.2.0 or later) and the independent controller which supports this function, the receiving card supports individual adjustment of red Gamma, green Gamma and blue Gamma, which can effectively control image non-uniformity under low grayscale and the visual experience. This function is easy to use and the adjustment takes effect immediately.

  In NovaLCT V5.2.0 or later, the adjustment can be performed without using or changing the video source.

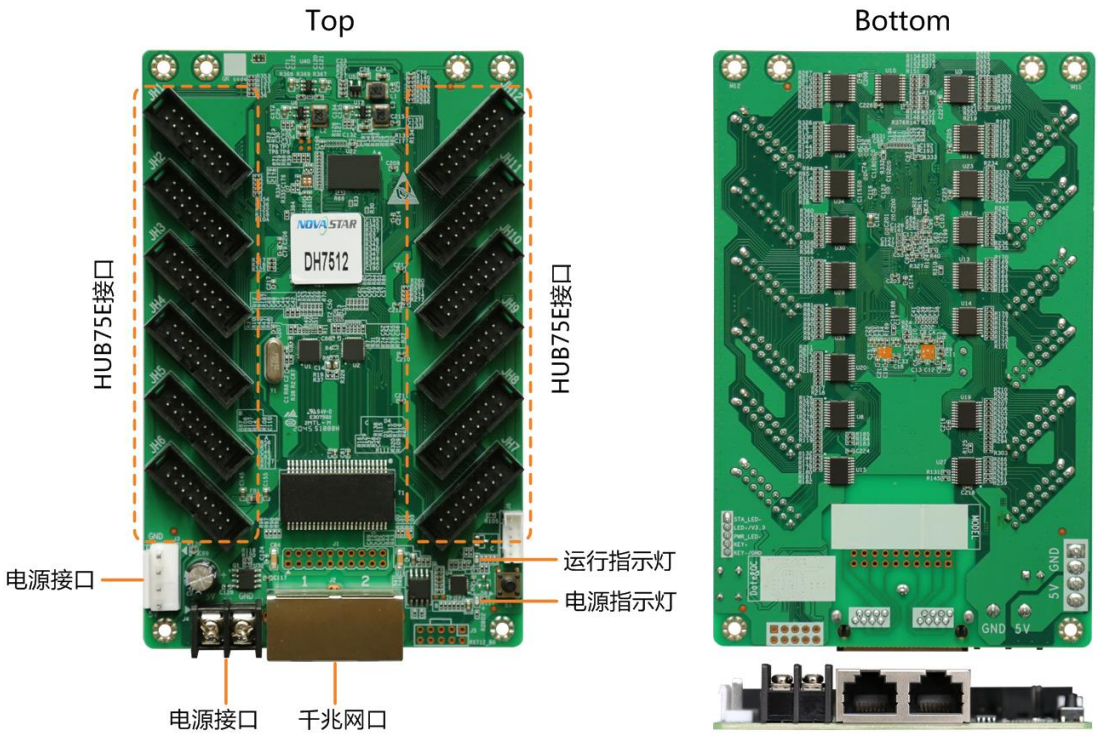
Improvements to Maintainability

- Mapping function
  The cabinets display the receiving card number and Ethernet port information, allowing users to easily obtain the locations and connection topology of receiving cards.
- Setting of a pre-stored image in receiving card
  The image displayed on the screen during startup, or displayed when the Ethernet cable is disconnected or there is no video signal can be customized.
- Temperature and voltage monitoring
  The temperature and voltage of the receiving card can be monitored without using peripherals.
- Cabinet LCD
  The LCD module connected to the cabinet can display the temperature, voltage, single run time and total run time of the receiving card.
- Bit error rate monitoring
  The Ethernet port communication quality of the receiving card can be monitored and the number of erroneous packets can be recorded to help troubleshoot network communication problems.
  NovaLCT V5.2.0 or later is required.
- Firmware program read back
  The firmware program of the receiving card can be read back and saved to the local computer.
  NovaLCT V5.2.0 or later is required.
- Configuration parameter read back
  The configuration parameters of the receiving card can be read back and saved to the local computer.

Improvements to Reliability

- Loop backup
  The receiving cards and the sending card form a loop via the main and backup line connections. If a fault occurs at a location of the lines, the screen can still display the image normally.
- Dual backup of configuration parameters
  The receiving card configuration parameters are stored in the application area and factory area of the receiving card at the same time. Users usually use the configuration parameters in the application area. If necessary, users can restore the configuration parameters in the factory area to the application area.
- Dual backup of the application program
  Two copies of the application program are stored in the receiving card at the factory to avoid the problem that the receiving card may get stuck due to program update exception.

Appearance

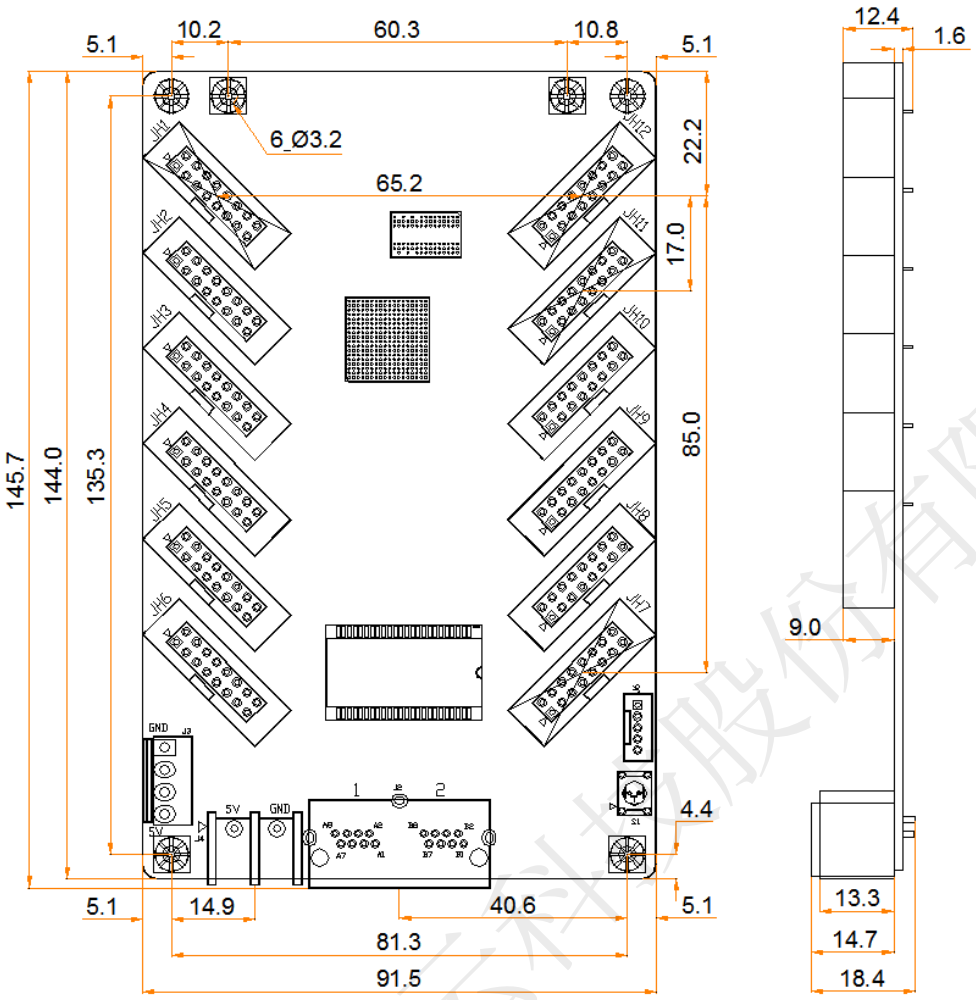


All product pictures shown in this document are for illustration purpose only. Actual product may vary.

Indicators

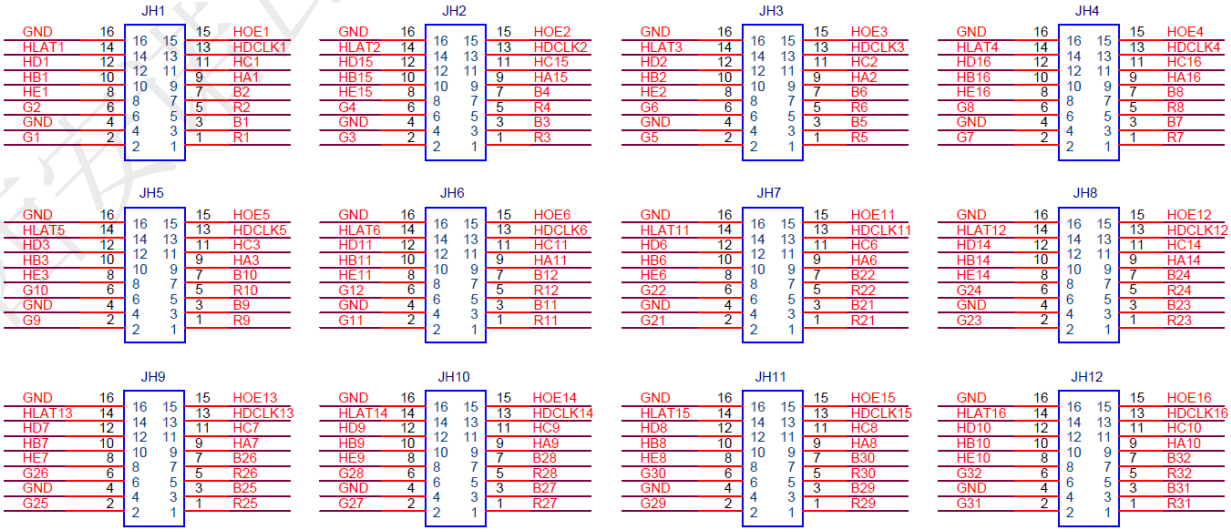
Indicator	Color	Status	Description
Running indicator	Green	Flashing once every 1s	The receiving card is functioning normally. Ethernet cable connection is normal, and video source input is available.
		Flashing once every 3s	Ethernet cable connection is abnormal.
		Flashing 3 times every 0.5s	Ethernet cable connection is normal, but no video source input is available.
		Flashing once every 0.2s	The receiving card failed to load the program in the application area and now is using the backup program.
		Flashing 8 times every 0.5s	A redundancy switchover occurred on the Ethernet port and the loop backup has taken effect.
Power indicator	Red	Always on	The power input is normal.

Dimensions



公差：±0.1　　单位：mm

Pins



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Specifications

Maximum Loading Capacity	512 ×512 pixels	
Electrical	Input voltage	DC 3.3 V to 5.5 V
Parameters	Rated current	0.5A
	Rated power consumption	2.5 W
Operating Environment	Temperature	–20°C to +70°C
	Humidity	10% RH to 90% RH, non-condensing
Storage Environment	Temperature	–25°C to +125°C
	Humidity	0% RH to 95% RH, non-condensing
Physical Specifications	Dimensions	145.7 mm ×91.5 mm ×18.4 mm
	Net weight	93.1g Note: It is the weight of a single receiving card only.
	Gross weight	12.9 kg Note: It is the total weight of the products, printed materials and packing materials packed according to the packing specifications.
Packing Information	Packing specifications	An antistatic bag and anti-collision foam are provided for each receiving card. Each packing box contains 40 receiving cards.
	Packing box dimensions	650.0 mm ×500.0 mm ×200.0 mm
Certifications	RoHS, EMC Class A	

The amount of current and power consumption may vary depending on many factors such as product settings, usage, and environment.

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